

Version-Aware Indexes for Machine-Annotated Text: Retrieval, Query Processing and Maintenance under Relabelling

Project outline for the PhD position in Developing Efficient Search Engines, NTNU Department of Computer Science (Jobbnorge 307690) · Asad Ikram

1. Problem and context

Classifiers now attach a sentiment, stance or toxicity label and a confidence score to each post or article. Analysts then ask for the most relevant posts that mention the budget, are labelled negative with confidence above 0.8 and appeared in the week after the announcement. They mix text relevance with label predicates and numeric ranges. Engines such as Lucene conjoin block-max WAND with a filter iterator, but block maxima ignore the predicate: when it is selective or anti-correlated with high-scoring terms, the top-k threshold rises slowly and most blocks are still decoded. And the annotations are not stable: a retrained classifier relabels part of the corpus, invalidating per-block score bounds and label-partitioned layouts, so an index built on old labels answers a different question.

The group's systems already index annotations as content: GYANI stores layers of part-of-speech, entity, time and number annotations (Gupta and Berberich, 2018); NASH answers range queries over temporal and numerical expressions on 365 million documents (Ohr and Gupta, 2024); KNITTIR adds operators that group, compare and rank entities (Vu and Gupta, 2024). These annotations are themselves tagger output, produced once and treated as fixed. This project extends that line to annotations carrying a confidence score and a model version. In my reading, filtered vector search treats labels as static (Gollapudi et al., 2023; Patel et al., 2024), real-time indexing adds documents rather than relabelling them (Busch et al., 2012), and time-travel text search versions documents, not annotations (Berberich et al., 2007; Anand et al., 2012).

2. Research questions

RQ1 (retrieval model). How should calibrated label probabilities and numeric scores be represented in an inverted index and combined with text relevance, so that queries mixing keywords, label predicates and score ranges return their top-k without post-filtering? Does ranking by expected relevance under label uncertainty beat hard thresholds?

RQ2 (efficient query processing). A posting block almost always holds every class of a three-way label, so bitmaps and confidence bounds prune only if docIDs cluster by label. Which docID ordering and summary placement (per term block or shared docID ranges) let block-max WAND skip safely, and at what selectivity and label-term correlation does this beat filter-iterator conjunction and columnar filtering?

RQ3 (maintenance under relabelling). When a new model version relabels part of the corpus, what is the cheapest way to keep answers exact for the latest version and for queries pinned to an earlier one: in-place patching, versioned delta segments merged in the background, or rebuilding? Can a cost model choose from the size, block spread and class pattern of the change?

RQ4 (modern hardware). Can predicates over bit-packed label identifiers and quantised confidences be evaluated with SIMD instructions fused with block-max skipping, rejecting blocks before their postings are decoded, and do patched blocks keep the alignment SIMD codecs need?

3. Theoretical perspectives

Safe dynamic pruning. WAND and block-max WAND are exact because stored upper bounds never underestimate a score (Broder et al., 2003; Ding and Suel, 2011). A predicate acts as a zero-or-one factor on the score, so a block whose bitmap or confidence interval cannot match has an upper bound of zero and is skipped safely; under RQ1's model, the bound is the block's maximum text score times its maximum label probability. Relabelling breaks safety asymmetrically: a change that raises a block's maximum makes its bound unsafe and must be applied before the next query; one that lowers it leaves the bound loose but safe, and can wait. RQ3 builds on this asymmetry and on amortised merge costs for indexes updated while queried (Lester et al., 2005).

Annotations as model-generated postings. Learned sparse retrieval already stores model output in inverted lists (Formal et al., 2021); guided traversal separates the signal that traverses from the one that scores (Mallia et al., 2022). Stable text postings can thus drive traversal while volatile annotation scores sit in a versioned column.

Labels as measurements. A calibrated confidence holds only for the model version that produced it, so the index records each label's version and version-pinned queries become time travel over models, not documents.

Working hypothesis H. Relabels concentrate near the decision boundary, so a confidence-banded layout isolates most changes in small segments. H assumes reasonable calibration and is testable on real model pairs; the cost model does not depend on it.

4. Methodological design

Systems. The query-processing core will extend PISA (Mallia et al., 2019), a C++ engine implementing WAND, block-max WAND, MaxScore and SIMD codecs (Lemire and Boytsov, 2015), keeping results comparable with the literature. At collection scale, the annotation layer could join the group's Java and HBase infrastructure.

Data and ethics. (a) MS MARCO (Bajaj et al., 2016) and BEIR (Thakur et al., 2021). (b) TREC Microblog Tweets2011 (Ounis et al., 2011), where it can still be rehydrated. (c) Common Crawl, as in the group's ICTIR 2024 work. (d) Public posts collected only through official APIs or licensed feeds under platform terms, with a Sikt-assessed data management plan; releases contain pseudonymised identifiers and derived labels, not text. No employer or client data is used.

Relabelling events. Each corpus is labelled by successive real models (encoder checkpoints, a lexicon, a BiLSTM, an LLM annotator, entity and temporal tagger releases), with scores calibrated by temperature scaling. Each event is characterised by how many labels change, their spread across blocks and between which classes, so results follow the shape of the change, not one model pair. Synthetic events (1 to 50 per cent changed, controlled skew) fill the gaps.

Workloads. Analyst templates: keyword plus label filter; keyword plus confidence range plus time window; KNITTIR-style aggregates; version-pinned queries. Selectivity (0.1 to 50 per cent) and label-term correlation are swept together.

Baselines. Exhaustive evaluation; post-filtering over block-max WAND; list intersection; per-label partitions; Lucene block-max WAND with a filter clause; a DuckDB-style columnar late filter, testing whether an annotation-aware index beats a join; filtered dense search (ACORN); full rebuild and append-only indexing for RQ3.

Metrics. Efficiency: mean and 99th-percentile latency, throughput, bits per posting, cost per relabelled document, and time from a new model version to a consistent index. Correctness: exactness of filtered and version-pinned top-k against exhaustive evaluation, needing no judgements. Effectiveness: existing qrels ignore labels, so for RQ1 I will human-label judged documents for a subset of MS MARCO and Microblog topics, add a small judged set of analyst queries, and report nDCG@10 and recall@1000.

Hardware. Single-thread and multi-core runs with AVX2 and, where available, AVX-512, SIMD on and off, on two CPU generations.

Risks. My C++ is the main risk, so semester 1 reproduces published PISA results before any algorithm changes. Pruning may gain little for unselective predicates; the sweep shows where it pays. If SIMD fusion gains little, RQ4 becomes a measurement section of the RQ2 paper. If API access narrows, the public collections serve every RQ.

5. Progress plan

Semester	Work	Output
1 (spring 2027)	Courses; TDT4117 material; literature review; C++ on PISA, reproducing block-max WAND; data management plan	Reproduced baselines; review chapter
2 (autumn 2027)	Courses; relabelling events and workloads; RQ1 representation and scoring; judgements begin	Event and workload generator
3 (spring 2028)	RQ2 pruning; selectivity sweeps; RQ4 SIMD fusion; 30 ECTS coursework completed; midway assessment	Benchmark paper (SIGIR resource track or CIKM)

4 (autumn 2028)	RQ2 and RQ4 evaluation; RQ3 versioned segments	Paper 1 (SIGIR or TheWebConf)
5 (spring 2029)	RQ3 cost model, version-pinned queries, test of H; cluster-scale runs	Paper 2 (VLDB or SIGMOD)
6 (autumn 2029)	RQ1 effectiveness study; code release; thesis	Paper 3 (TheWebConf or CIKM); thesis
Four-year option	Teaching assistance (Databases, Data Mining, Information Retrieval) at 25 per cent throughout, with the research plan above spread over four years	Teaching record

6. Fit and preparation

At ArtemisAI, where I am founder and CTO, our fine-tuned encoder classifiers, retrained through several versions, label our posts and a subset of the 3.39 million stored comments in weekly batches. The evaluation reference moved with them: v1 to v3 were scored against a 97-post human holdout, v4 onwards against LLM labels. The team also saw one annotator's positive share fall from 99.8 to 55.4 per cent with nothing changed on our side. An index over such labels must know which model produced them.

My MSc in Data Analytics (Distinction, project mark 78) labelled about 279,000 posts with a lexicon, a BiLSTM and a fine-tuned BERT, calibrated by temperature scaling and validated against human coding (Cohen's kappa about 0.78). A methods manuscript with Dr Ella Haig is complete but not yet submitted.

At Dubizzle Labs I redesigned a data warehouse, raising ETL throughput by 60 per cent; at M+C Saatchi Fluency I chose a composite key so that a 770,000-row weekly pipeline re-translates only unseen records: the reuse question RQ3 asks inside an index.

I have not yet built an inverted index; my production code is Python, SQL and PySpark. The plan front-loads C++ and TDT4117, which also prepares me to assist in teaching it.

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